
Algorithm 2 PySR search with four evolvable operator slots

Input Data $D = (X, y)$; expression grammar; search budget; populations $P_1, \dots, P_J$; size limit $c_{\max}$; tournament size k ; crossover probability $p_{\times}$.

```
1 Initialize and score random expressions; initialize hall of fame  $H$  and size frequencies  $f$ 
2 while search budget and iteration allowance remain do
3   Choose a population  $P$  to evolve    # Population workers may run concurrently
4   for each inner cycle, with scheduled temperature  $T$  do
5     for  $e = 1, \dots, \lceil |P|/k \rceil$  do
6       if  $\text{Unif}(0, 1) < p_{\times}$  then
7          $a, b \leftarrow \text{Select}(P, f), \text{Select}(P, f)$ 
8          $C \leftarrow \text{CONSTRAINEDSUBTREECROSSOVER}(a, b)$ 
9         Score both children using Loss $(\cdot, D)$  and the cost transform
10      else
11         $a \leftarrow \text{Select}(P, f)$ 
12        Draw a feasible mutation from the weighted menu, including Mutate
13         $C \leftarrow \text{PROPOSEANDACCEPT}(a, T, f, c_{\max}, \text{Loss})$ 
14      if the proposal succeeds then
15         $I \leftarrow \text{DISTINCTVICTIMS}(P, |C|, \text{Survive})$ 
16      Replace members at  $I$  with  $C$ , assigning offspring birth times and lineage
17  Retain the best expression at each complexity in a local hall of fame
18  Simplify expressions; probabilistically optimize constants; refresh scores
19  Merge local and population champions into  $H$ ; update size frequencies  $f$ 
20  If enabled, inject migrants from population elites and the Pareto frontier of  $H$ 
21  Advance the size-limit schedule and the frequency window
22 return the loss–complexity Pareto frontier of  $H$ 
```

Bold sans-serif marks an evolvable operator slot. When supplied, the mutation slot adds a move to PySR’s weighted menu (tree edits, constant changes, simplification, and other configured moves); selection, survival, and loss replace their respective hooks. **Survive** returns a replacement index, not a new population. For crossover, both victims are chosen before insertion, with the first index excluded from the second call.

Baseline hooks. Loss is the configured pointwise regression loss (MSE by default); cost normalizes loss and adds parsimony. Selection ranks a size- k sample by cost, optionally multiplied by $\exp(\lambda f_c)$, and draws rank r with probability proportional to $p(1 - p)^{r-1}$. Survival removes the oldest eligible member.

Acceptance. Ordinary tree mutations pass constraints and a non-NaN cost check, then optional annealing and size-frequency tests. Their acceptance probability is $\min\{1, \exp[-(C_{\text{child}} - C_{\text{parent}})/(\alpha T)] f_{c_{\text{parent}}} / f_{c_{\text{child}}}\}$ when both options are enabled. Crossover uses constraints without this test. Some built-in moves return immediately. The default failure policy skips unsuccessful proposals.
